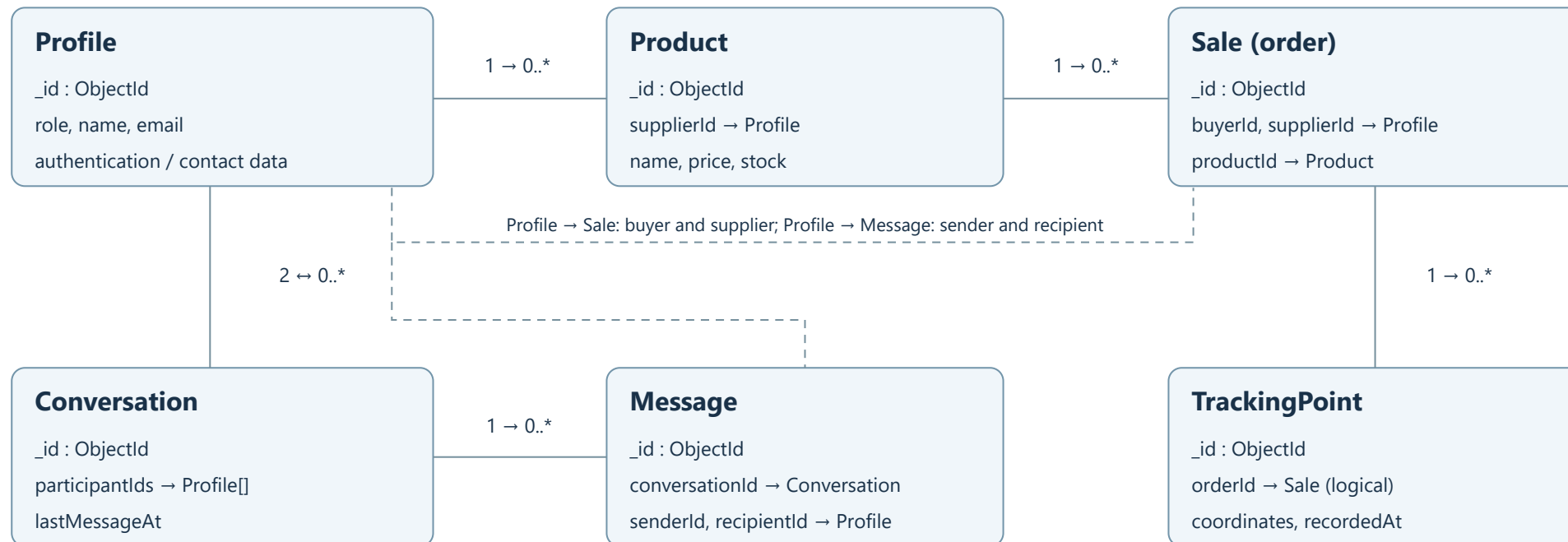


## Database design • Entity relationships

MongoDB document storage through Mongoose. Lines show intended application relationships; ObjectId references are not database-enforced foreign keys.



### Cardinality key

1 → 0..\*: one parent may have many children. Each order has one buyer, one supplier and one product. Each message has one sender and one recipient. Dashed links show these additional Profile references.

### Conversation and independent records

The API creates conversations with two opposite-role participants; the schema stores an array. A profile may join many conversations. Feedback is an independent collection with no ObjectId relationship.

# Data dictionary • Accounts, catalog and chat

All models have an automatic ObjectId \_id. All except TrackingPoint enable createdAt and updatedAt. R = required by the declared schema; U = unique index declaration.

| Field(s)                                                     | Type / constraint                         | Purpose                                                         |
|--------------------------------------------------------------|-------------------------------------------|-----------------------------------------------------------------|
| <b>Profile:</b> role; name; email                            | String R; String R; String R/U            | Role enum: business, supplier. Email lowercased and trimmed.    |
| fullName; phone; town; about; category                       | String (optional)                         | Identity, location and profile information.                     |
| deliveryInfo; minimumOrder; businessHours; profilePhotoUrl   | String (optional)                         | Supplier details and hosted profile image.                      |
| facebookId; googleId                                         | String; U, sparse                         | Optional provider identities.                                   |
| passwordHash; passwordResetTokenHash; passwordResetExpiresAt | String; String; Date; excluded by default | Password verification and reset credentials.                    |
| <b>Product:</b> name; supplierId                             | String R; ObjectId R → Profile            | Product name and owning supplier.                               |
| category; price; stock                                       | String; Number ≥ 0; Number ≥ 0, default 0 | Classification, price and available inventory.                  |
| images; gcashQrUrl                                           | String[] default []; String default empty | Cloudinary image URLs and optional GCash QR URL.                |
| <b>Conversation:</b> participantIds; lastMessageAt           | ObjectId[] → Profile; Date                | Participant references (required per element); recent activity. |
| <b>Message:</b> conversationId; senderId; recipientId        | ObjectId R → Conversation / Profile       | Conversation membership and message parties.                    |
| text; readAt                                                 | String R, trimmed; Date optional          | Message content and recipient read timestamp.                   |

**Validation boundary:** required schema fields and indexes are distinct from route rules. For example, roles are checked by the API when creating products and conversations; a Profile reference alone does not enforce a supplier role.

## Data dictionary • Orders, tracking and feedback

| Field(s)                                                                 | Type / constraint                                    | Purpose                                                        |
|--------------------------------------------------------------------------|------------------------------------------------------|----------------------------------------------------------------|
| <b>Sale:</b> supplierId; buyerId; productId                              | ObjectId R → Profile / Product                       | Order parties and purchased product.                           |
| quantity; total                                                          | Number R ≥ 1; Number R ≥ 0                           | Ordered amount and order total.                                |
| deliveryAddress; deliveryTown; contactPhone; notes                       | String, trimmed                                      | Delivery and buyer instructions.                               |
| status; soldAt                                                           | String enum; Date default now                        | pending / confirmed / completed / cancelled; sale timestamp.   |
| paymentMethod                                                            | String; default Cash on Delivery (COD)               | New-order route accepts COD and GCash.                         |
| payment.status                                                           | String enum                                          | cod / pending / approved / rejected.                           |
| payment.gcashName; gcashPhone; gcashReference; qrUrl                     | String                                               | Payer details, reference and QR snapshot URL.                  |
| payment.proof; proofType                                                 | Buffer, excluded by default; String                  | Private receipt bytes and media type.                          |
| payment.submittedAt; reviewedAt; reviewNote                              | Date; Date; String                                   | Submission, review time and rejection/review reason.           |
| <b>TrackingPoint:</b> orderId; latitude; longitude; accuracy; recordedAt | ObjectId R; Number; Number; Number; Date default now | Logical order link and reported GPS fix. No timestamps option. |
| <b>Feedback:</b> developerEmail; name; email; message                    | String R for all fields                              | Independent feedback. Emails lowercase; sender fields trimmed. |

**Embedded payment:** the payment object belongs to Sale; there is no separate Payment collection. **Derived tracking number:** CD- plus uppercase order ID is added during serialization; it is not a stored schema field.

Payment subfield names after the first field remain under payment. Tracking coordinates are validated by the location route in addition to their basic Number schema types.

# Indexes, integrity and storage lifecycle

| Model                   | Declared indexes beyond the automatic <code>_id</code> index                        |
|-------------------------|-------------------------------------------------------------------------------------|
| Profile                 | role; town; email (unique); facebookId (unique, sparse); googleId (unique, sparse). |
| Product                 | name; category; supplierId.                                                         |
| Sale                    | supplierId; soldAt.                                                                 |
| Message                 | conversationId.                                                                     |
| TrackingPoint           | orderId.                                                                            |
| Conversation / Feedback | No additional indexes explicitly declared in their schemas.                         |

## Transactional order rules

- Confirmation atomically deducts available product stock and updates order status using a MongoDB transaction.
- Cancelling a confirmed order restores its stock. Pending cancellation does not change stock.
- Closed orders cannot reopen; completion requires confirmation. GCash approval gates confirmation and completion.
- MongoDB Atlas or another replica set is required for these transaction paths.

## Media and deletion behavior

- Cloudinary stores product, QR and profile images; documents retain their URLs.
- Payment receipts are stored in Sale as binary buffers, excluded from ordinary projections and order JSON.
- Account deletion explicitly removes related messages, conversations, sales, products and the profile.
- The inspected deletion route does not remove TrackingPoint records or Cloudinary assets. MongoDB references do not cascade automatically.

**Design scope:** this document describes the current schema and route behavior. Declared indexes were read from code; their presence in a live database was not inspected. Future database changes should address orphan cleanup and query-driven indexing separately.

Sources: backend/server.js (schemas, order routes, receipt access and account deletion); backend/order-status.js; backend/payment.js; PAYMENTS.md; TRACKING.md. Collection names follow Mongoose model naming defaults: profiles, products, sales, conversations, messages, trackingpoints and feedbacks.